AURA

ASSET UNIFIED REASONING AGENT · PROBLEM 08

The Unified Asset & Operations Brain

An AI that connects the knowledge scattered across an industrial plant — engineering, maintenance, reliability, safety, compliance and email — into one queryable brain, and surfaces what no single team can see.

WORKING PROTOTYPE · KNOWLEDGE GRAPH · CITED RAG · AGENTIC COMPLIANCE

Submission Document · Industrial Knowledge Intelligence

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01 — EXECUTIVE SUMMARY

One reasoning agent across every asset

A typical large plant runs across 7–12 disconnected document systems. The information needed to run, maintain and certify an asset already exists — it is simply scattered, and no person or tool sees all of it at once. **AURA (Asset Unified Reasoning Agent)** ingests every document type, links them into a self-updating knowledge graph, and answers operational, maintenance, engineering and compliance questions with cited, confidence-scored intelligence — on any device.

The result is not a better search box. It is a reasoning layer that connects maintenance history, sensor data, manufacturer manuals, safety reports, regulations and email into a single picture — exposing recurring failures, compliance gaps and safety risks that fragmentation keeps hidden. This document describes the problem, the solution, a working prototype, the demonstration scenario, the architecture, and how AURA maps to the evaluation criteria.

35%
of an engineer's hours lost searching for or recreating documents that already exist.
MCKINSEY, 2024

18–22%
of unplanned downtime in Indian heavy industry traced to knowledge fragmentation.
BIS RESEARCH

25%
of experienced engineers retire within a decade — taking undocumented know-how with them.
NASSCOM–EY

02 — THE PROBLEM

Knowledge is everywhere. Insight is nowhere.

Asset-intensive industries store their operational knowledge across many incompatible systems: P&IDs and engineering drawings in one repository, maintenance work orders in a CMMS, operating procedures in a document store, inspection and sensor data in a reliability tool, and regulatory submissions scattered across email archives. Each system is individually fine; collectively they prevent anyone from seeing the whole picture of an asset.

This fragmentation is not an inconvenience — it is a measurable operational liability:

A safety problem

Maintenance and operating decisions get made without complete equipment history or failure-pattern context — until something leaks, trips, or burns.

A quality & compliance problem

Required investigations and the evidence to prove them sit in separate systems. Gaps surface during an audit, not before it.

An efficiency problem

The same failures recur, the same questions get re-answered from scratch, and unplanned downtime compounds over time.

A knowledge-cliff problem

As experienced staff retire, decades of undocumented judgement leave with them. Once gone, it cannot be recovered.

> **The thesis**
> This is a connection problem, not a storage problem. Whoever connects the knowledge first gains a structural advantage in how they operate, maintain and improve their assets.

AURA — a unified reasoning layer over every document

AURA turns a plant's heterogeneous document estate into one queryable brain through four capabilities:

Capability	What it does
Ingest anything	PDFs, P&IDs, scanned forms, spreadsheets and email — structured and unstructured — through pluggable connectors.
Connect everything	Extracts entities and relationships into a knowledge graph linking assets, failures, procedures, regulations and people.
Answer with citations	A retrieval-augmented copilot that answers any question, cites its sources, and shows a confidence level on every reply.
Act with agents	Purpose-built agents for root-cause analysis, compliance-gap detection, and lessons-learned pattern mining.

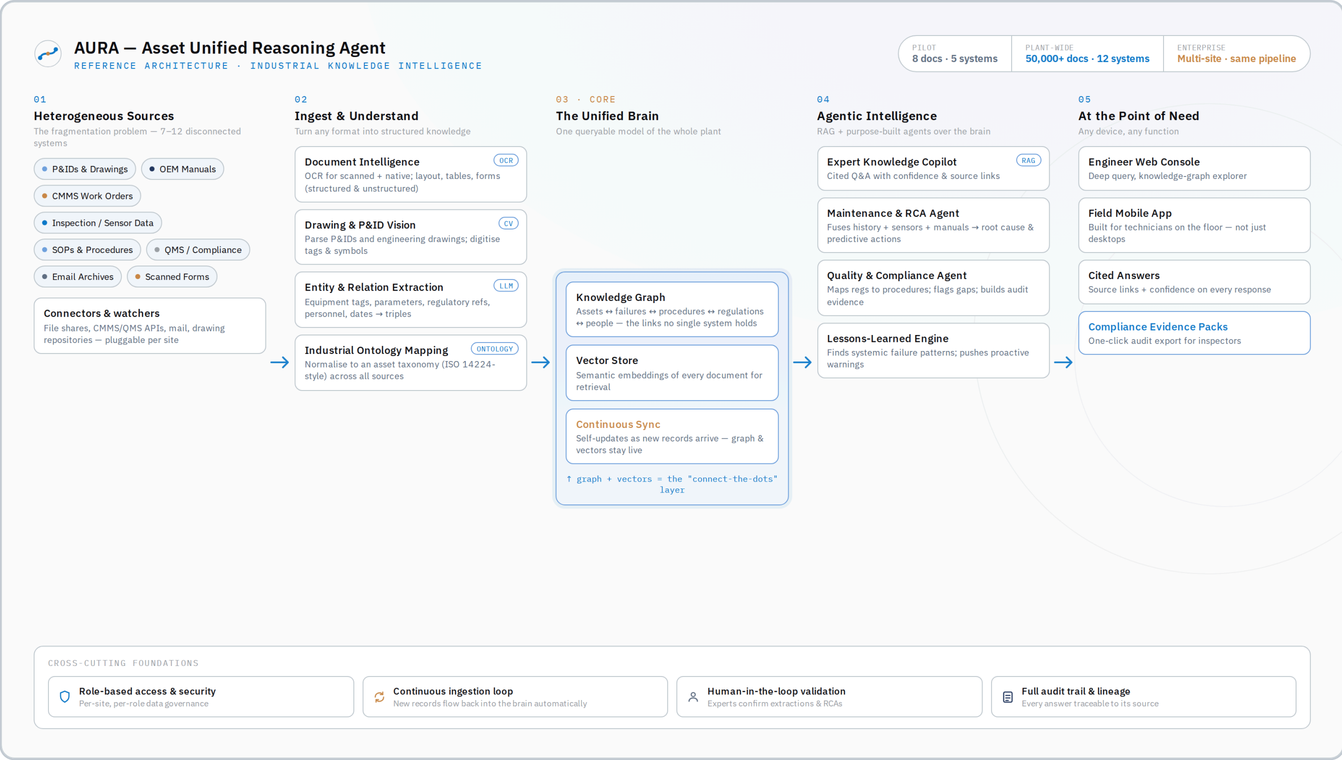
Coverage of the problem statement's solution areas

AURA addresses all five illustrative directions in Problem 08:

- **Universal Document Ingestion & Knowledge Graph** — the ingestion + entity-extraction + graph pipeline (core of AURA).
- **Expert Knowledge Copilot** — the cited, confidence-scored RAG assistant, built mobile-first for field technicians.
- **Maintenance Intelligence & RCA Agent** — fuses work orders, sensor data and manuals to surface root cause and predictive actions.
- **Quality & Regulatory Compliance Intelligence** — maps regulations to current procedures and records, flagging gaps and assembling audit evidence.
- **Lessons-Learned & Failure Intelligence** — mines incident and near-miss history for systemic patterns and proactive warnings.

04 — HOW IT WORKS

One pipeline. Five layers. Any scale.



AURA reference architecture — sources → ingest & understand → the unified brain → agentic intelligence → point of need.

1 · Sources. Pluggable connectors and watchers pull from file shares, CMMS/QMS APIs, mail and drawing repositories.

2 · Ingest & understand. OCR and document intelligence handle scanned and native files; computer vision parses P&IDs and drawings; an LLM extracts entities and relationships; an industrial ontology normalises everything to an asset taxonomy.

3 · The unified brain. A knowledge graph plus a vector store form the connect-the-dots layer, kept live by continuous sync as new records arrive.

4 · Agentic intelligence. RAG and purpose-built agents (copilot, RCA, compliance, lessons-learned) reason over the brain.

5 · Point of need. A web console for engineers, a mobile app for field technicians, cited answers, and one-click compliance evidence packs.

Cross-cutting: role-based access & security, a continuous ingestion loop, human-in-the-loop validation, and a full audit trail with data lineage.

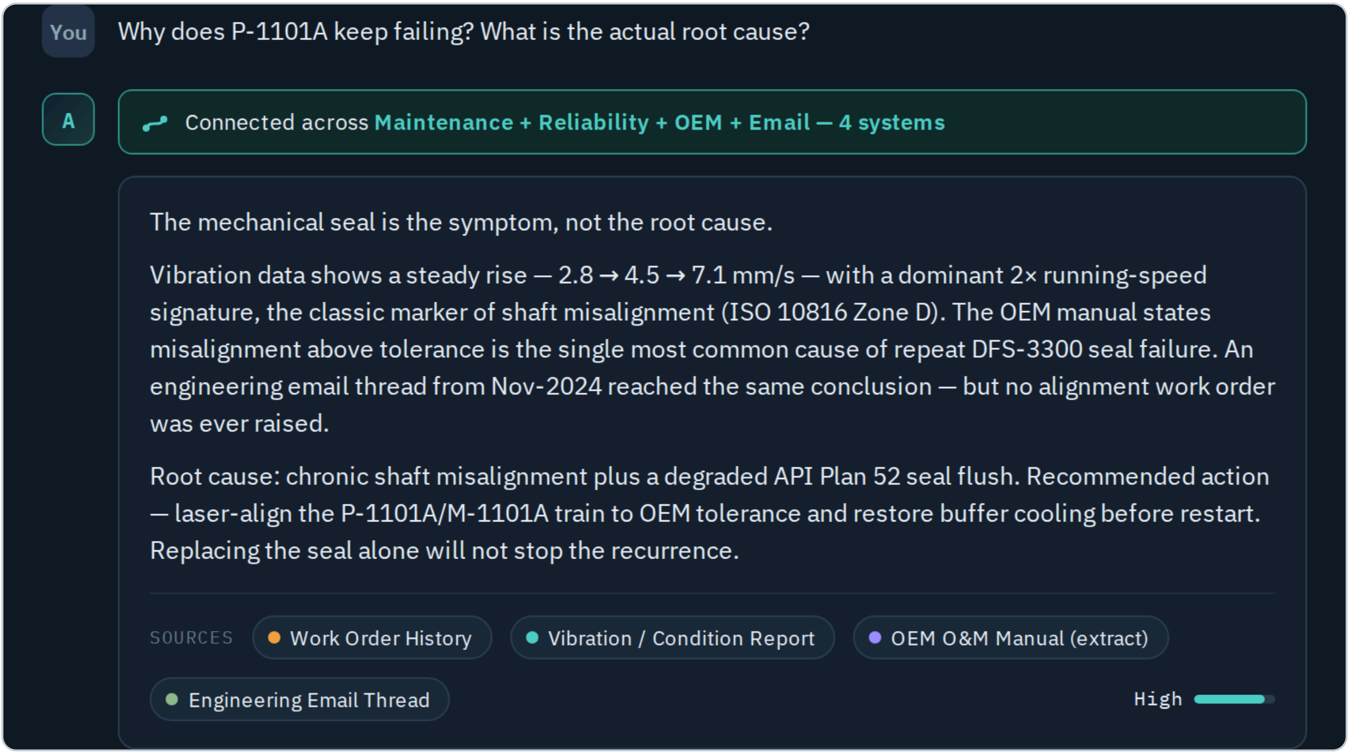
Technology

RAG · Knowledge Graph · Industrial Ontology · Computer Vision (P&ID) · OCR & Doc Intelligence · Vector Embeddings · Agentic AI · QMS Integration

05 — THE WORKING PROTOTYPE

A live copilot, not slideware

The prototype is a functioning Industrial Knowledge Copilot. In **Live AI** mode it performs real retrieval-augmented generation over the indexed document set, so it answers novel questions a judge types. A fully offline **Demo** mode provides instant, pre-validated answers for reliable recording. Every answer carries source citations and a confidence level, and a knowledge-graph view visualises the links between documents.



AURA connects four systems to answer a root-cause question — with source citations and a confidence score.

The copilot is demonstrated through six representative questions spanning recall, root-cause reasoning, engineering-spec retrieval, compliance-gap detection, field safety, and graph exploration — each returning cited answers grounded in the documents.

06 — DEMONSTRATION SCENARIO

One pump. Three failures. Five systems. Zero connections.

The prototype runs on a realistic corpus centred on one critical asset at a fictional refinery — Crude Charge Pump **P-1101A**. Across 2024 the pump's mechanical seal failed three times (March, July, November) and produced a hydrocarbon-seepage near-miss in December. Each event was handled in a different system:

System	What it held
CMMS	Three work orders — each closed as an isolated seal replacement.
Reliability	Vibration rising 2.8 → 4.5 → 7.1 mm/s with a 2× signature — textbook misalignment.
OEM manual	Misalignment is the #1 cause of repeat seal failure for this seal type.
Email	Two engineers diagnosed it correctly — but no work order was ever raised.
Compliance	A root-cause analysis was statutorily required, and never done.

Asked *"why does P-1101A keep failing?"*, AURA fuses all five and concludes the real root cause is **chronic shaft misalignment plus a degraded seal-flush** — not the seal being replaced — and recommends laser alignment before restart. The knowledge graph makes the missed link visible at a glance.

The knowledge graph: selecting the hidden root cause lights up the four systems that each held one piece of the puzzle.

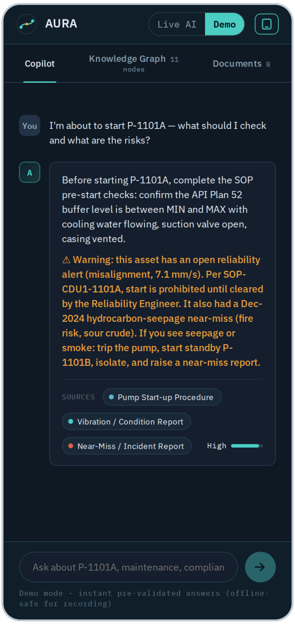
> **Why this matters**
>
> Every clue existed — in a different system. No single person connected them. AURA does, in seconds, with every claim traceable to its source.

07 — BUSINESS IMPACT & SCALABILITY

Hours, not weeks. Incidents prevented. Audits passed.

By connecting what fragmentation hides, AURA converts repeated, reactive work into single, root-cause fixes:

- **3 → 1:** three repeat failures collapse into one root-cause correction.
- **1 near-miss avoidable:** a proactive alert flags the open risk before restart.
- **Minutes, not weeks** to a cited answer that previously meant searching five systems.
- **1-click audit evidence** versus days of manual assembly.



Mobile field view — proactive safety warning

Scalability — the same pipeline at any size

Stage	Scope
Pilot	8 documents · 5 systems · one critical asset — proven end-to-end.
Plant-wide	50,000+ documents · 12 systems · every asset across a site.
Enterprise	Multi-site, multi-plant — one brain across the organisation.

Scaling means adding a connector, not rebuilding: the ingestion → graph → RAG → agent pipeline is identical at every stage.

08 — WHY AURA & EVALUATION ALIGNMENT

Not a better search box. A brain that connects the dots.

- **Cross-system intelligence** — surfaces patterns no single team or tool can see.
- **Trustworthy by design** — citations, confidence scores and a full audit trail on every answer.
- **Field-first** — built for technicians on the floor, not just engineers at desks.
- **Agentic compliance** — automatically detects gaps and assembles audit evidence.

Criterion	Weight	How AURA delivers
Innovation	25%	Surfaces a hidden cross-system root cause and an agentic compliance gap that fragmentation keeps invisible.
Business Impact	25%	Downtime avoided, a near-miss prevented, audit risk closed, time-to-answer cut from weeks to seconds.
Technical Excellence	20%	Live RAG + knowledge graph + CV/OCR + agents, with citations and confidence — a working prototype.
Scalability	15%	One pipeline from pilot to enterprise; scale by adding connectors.
User Experience	15%	Clean cited answers, a visual knowledge graph, and a mobile field experience.

Evaluation focus

AURA targets the stated measures directly: entity-extraction across document types, query-answer quality on expert questions, knowledge-graph linkage completeness, time-to-answer versus traditional search, and compliance-gap detection — demonstrated on a realistic industrial document set.

From prototype to production

Next steps

- Connector SDK for common CMMS / QMS / historian systems.
- Hardened P&ID computer-vision pipeline for tag and symbol extraction.
- Predictive maintenance scoring on top of the RCA agent.
- Role-based security, SSO and per-site data governance.
- Human-in-the-loop review queue for extractions and RCAs.

Deliverables in this submission

- This detailed document (PDF).
- Pitch deck (PPTX).
- Working prototype (interactive web app).
- Reference architecture diagram.
- 3–4 minute demo video.

Links

Resource	Link
GitHub repository	[paste your GitHub URL]
Demo video	[paste your video / Google Drive link]
Live prototype	[optional hosted link, if deployed]